Practical 2 — Superstore order size

Numerical summaries, distributions and unusual orders · 307312 Applied Statistics

- Companion to centre, spread, percentiles and boxplots in Part 2.
- Continue from Practical 1.
- Allow two sessions of about 40–50 minutes.

By the end, you should be able to:

- Aggregate transactions at the correct level.
- Explain the difference between typical order size and total revenue.
- Describe skewness, spread and percentile thresholds.
- Investigate unusual records while preserving the complete analysis.

1. The operations manager's question

> What is a typical order worth, and how much variation must operations accommodate?

- **Unit:** one order after combining all its product lines.
- **Source:** Tableau's fictitious Superstore, downloaded 6 November 2026.
- **Scope:** United States; orders placed in 2023–2025.
- **Records:** 3,322 orders built from 6,682 product lines.
- **Amounts:** recorded dollars; `sales` is the line/order sales field, not a list price.
- **Limit:** these describe the course extract, not actual future sales.

2. Import and aggregate

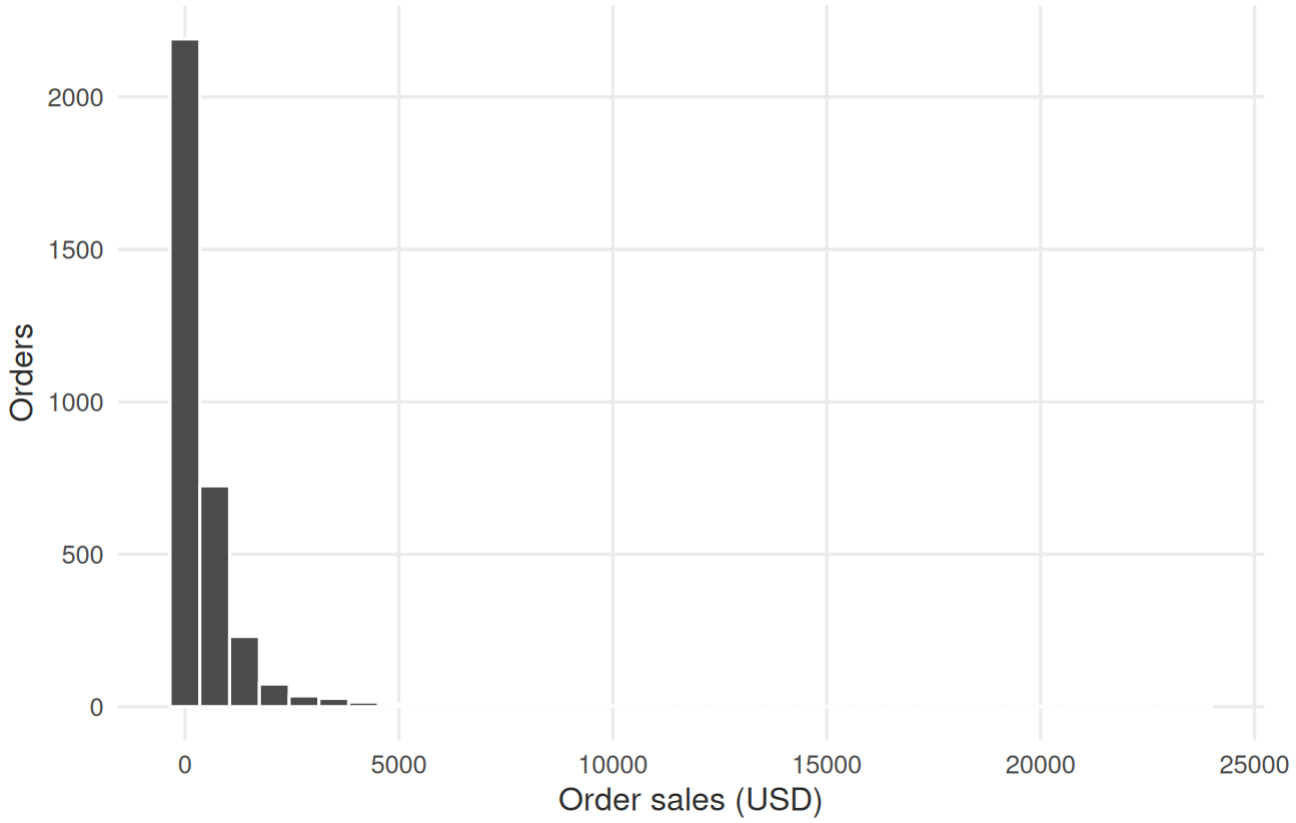
```
data_dir <- if (dir.exists("data")) "data" else if (dir.exists(file.path("..", "data")))
        file.path("..", "data") else file.path("..", "..", "data")
source(file.path(data_dir, "..", "lessons", "_includes", "course-tools.R"))
lines <- read.csv(file.path(data_dir, "business", "superstore-lines.csv"),
                  stringsAsFactors = FALSE, colClasses = c(postal_code = "character"))
orders <- make_orders(lines)
stopifnot(nrow(orders) == n_distinct(lines$order_id),
          isTRUE(all.equal(sum(orders$sales), sum(lines$sales))))
knitr::kable(head(orders[c("order_id", "sales", "profit", "quantity", "line_count")] ), digits = 2)
```

order_id	sales	profit	quantity	line_count
US-2023-100006	377.97	109.61	3	1
US-2023-100090	699.19	-19.09	9	2
US-2023-100279	22.38	10.74	2	1
US-2023-100293	91.06	31.87	6	1
US-2023-100328	3.93	1.33	1	1
US-2023-100363	21.38	7.72	5	2

- `data/business/superstore-lines.csv`; guide: `data/business/README.md`.
- Sales, profit and quantity are **summed** across an order's lines.
- Do not average line sales to estimate average order sales.
- The reconciliation checks that aggregation neither loses nor duplicates revenue.

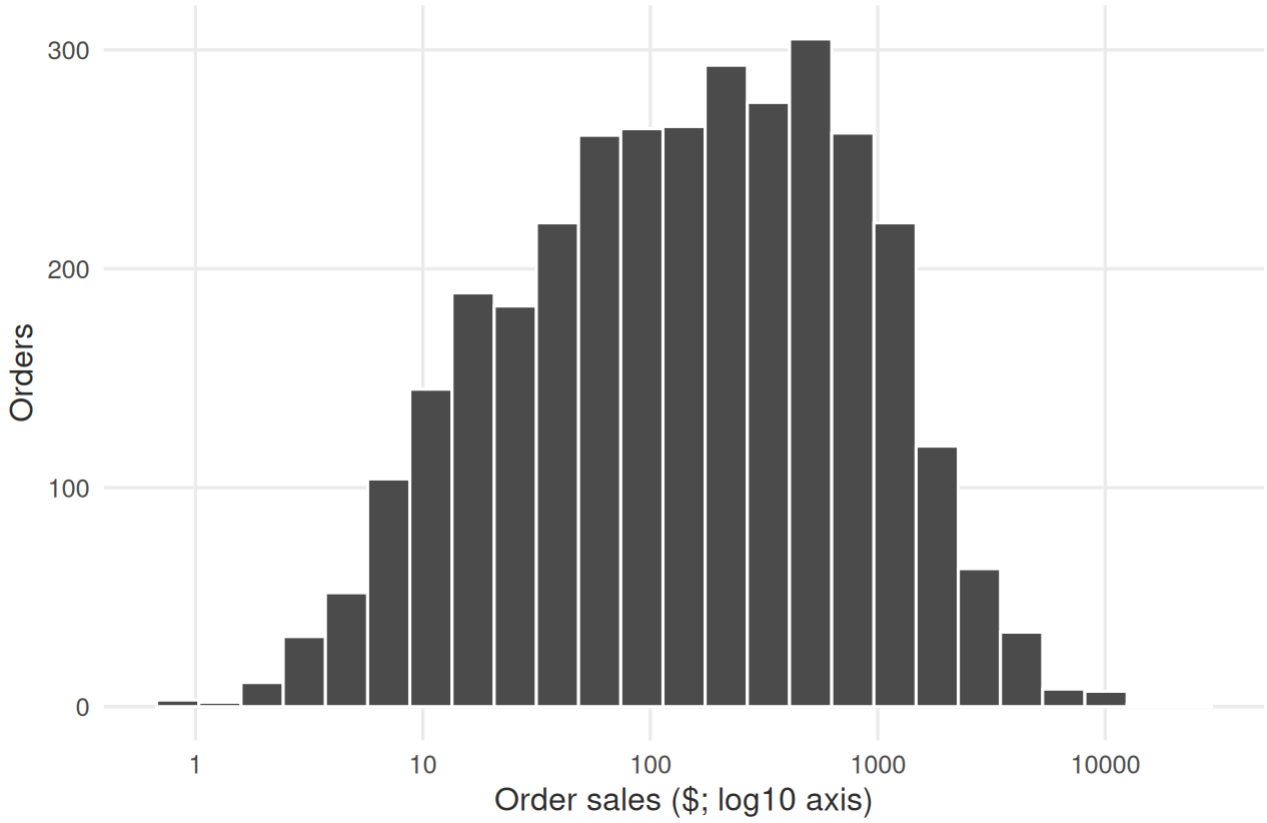
3. Look at the distribution first

```
ggplot(orders, aes(sales)) +
  geom_histogram(bins = 35, fill = petra_ink, colour = "white") +
  labs(x = "Order sales (USD)", y = "Orders")
```



- Most orders are relatively small; a few are very large.
- The long right tail affects the mean and SD.
- A histogram's bin width changes the visible detail.
- A tall bar shows many orders in an interval, not many dollars of revenue.

```r
ggplot(orders, aes(sales)) +
  geom_histogram(bins = 25, fill = petra_ink, colour = "white") +
  scale_x_log10() + labs(x = "Order sales ($; log10 axis)", y = "Orders")
```



- The log axis makes the lower range easier to inspect without deleting large orders.
- Equal spacing on this axis means equal ratios, not equal dollar differences.

4. Choose useful numerical measures

```r
x <- orders$sales
summary_table <- data.frame(
  statistic = c("Orders", "Mean (USD)", "Median (USD)", "Sample SD (USD)",
                "Minimum (USD)", "Q1 (USD)", "Q3 (USD)", "Maximum (USD)", "IQR (USD)"),
  value = c(length(x), mean(x), median(x), sd(x), min(x),
            quantile(x, .25, type = 7), quantile(x, .75, type = 7), max(x), IQR(x)))
knitr::kable(summary_table, digits = 2)
```

statistic	value
Orders	3322.00
Mean (USD)	470.80
Median (USD)	153.98
Sample SD (USD)	982.97
Minimum (USD)	0.84
Q1 (USD)	38.71
Q3 (USD)	535.34
Maximum (USD)	23661.23
IQR (USD)	496.62

- Mean order sales: about **$471**.
- Median order sales: about **$154**.
- Mean explains revenue per order; median describes the middle order.
- SD is sensitive to the right tail; IQR describes the middle half.
- `sd()` uses denominator $n - 1$; the frame SD uses N and is slightly smaller here.

Revenue reconciliation

```
c(total_sales = sum(x), mean_times_orders = mean(x) * length(x),
  median_times_orders = median(x) * length(x))
```

```
       total_sales   mean_times_orders median_times_orders
         1563985.6           1563985.6            511528.2
```

- Mean × number of orders reproduces total revenue.
- Median × number of orders generally does not.
- The median is not an alternative way to calculate the total.

Absolute and relative spread

```
c(range_dollars = diff(range(x)),
  mean_absolute_deviation = mean(abs(x - mean(x))),
  coefficient_of_variation_pct = 100 * sd(x) / mean(x))
```

```
              range_dollars      mean_absolute_deviation
                  23660.3920                    501.6800
coefficient_of_variation_pct
                    208.7897
```

- Mean absolute deviation above is not R's `mad()`, which uses the median and a scaling constant.
- CV is appropriate here because sales have a meaningful zero and positive mean.
- Do not apply the same CV rule to profits that can be negative or near zero on average.

5. Percentiles for operational planning

> What order values define the upper 10% and upper 5% of the observed distribution?

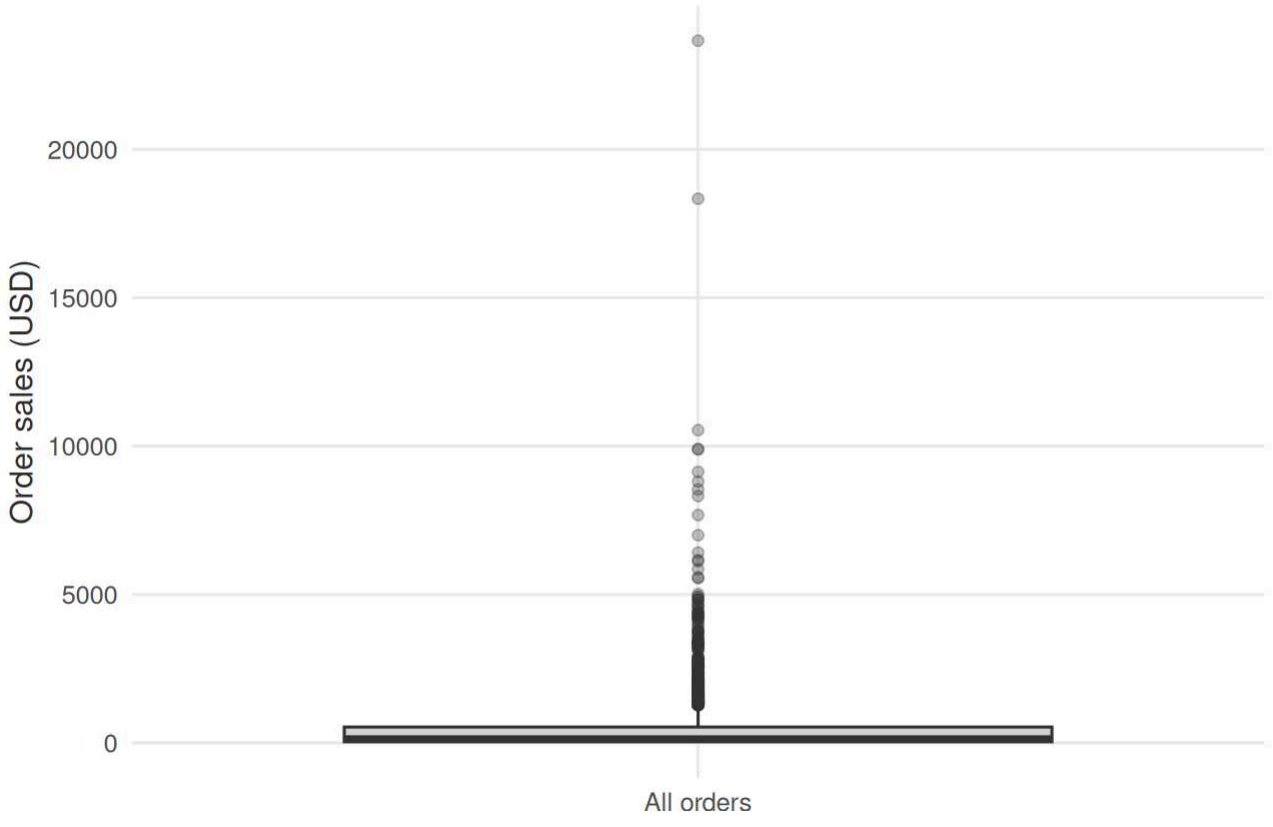
```
knitr::kable(data.frame(percentile = c("50th", "75th", "90th", "95th", "99th"),
  order_sales = as.numeric(quantile(x, c(.5, .75, .9, .95, .99), type = 7))), digits = 2)
```

percentile	order_sales
50th	153.98
75th	535.34
90th	1190.81
95th	1799.67
99th	4277.10

- Percentiles describe positions in the observed distribution.
- Interpolation and ties mean a threshold may not leave exactly the named percentage strictly above it.
- A sales percentile does not directly measure parcel size, picking time or staffing needs.
- Additional operational variables would be needed for those decisions.

6. Boxplots and unusual orders

```
ggplot(orders, aes(x = "All orders", y = sales)) +
  geom_boxplot(fill = petra_accent, outlier.alpha = .3) +
  labs(x = NULL, y = "Order sales (USD)")
```



```r
quartiles <- quantile(x, c(.25, .75), type = 7)
lower_fence <- quartiles[1] - 1.5 * IQR(x)
upper_fence <- quartiles[2] + 1.5 * IQR(x)
flagged <- orders %>% filter(sales < lower_fence | sales > upper_fence)
c(lower_fence = lower_fence, upper_fence = upper_fence, flagged_orders = nrow(flagged))
```

```
lower_fence.25% upper_fence.75%  flagged_orders
      -706.2188       1280.2713        301.0000
```

- Flags identify records worth investigating.
- The source contains genuine large recorded transactions; a flag alone does not establish an error.
- Boxplot hinges and reported quartiles can differ slightly for some sample sizes.

```r
knitr::kable(orders %>% arrange(desc(sales)) %>%
  select(order_id, sales, profit, line_count, quantity) %>% head(5), digits = 2)
```

order_id	sales	profit	line_count	quantity
US-2023-145317	23661.23	-1789.73	7	25
US-2025-118689	18336.74	8762.39	5	18
US-2023-139892	10539.90	-1878.79	7	37
US-2023-116904	9900.19	4668.69	4	17
US-2025-117121	9892.74	4946.37	1	13

- Inspect the underlying product lines before deciding what a large order means.
- A large order may contain one costly product or several products.
- Keep all records in the main summary.

A labelled sensitivity comparison

```r
central_orders <- orders %>% filter(sales >= lower_fence, sales <= upper_fence)
knitr::kable(data.frame(
  scope = c("Full frame — primary result", "Within IQR fences — sensitivity only"),
  n = c(nrow(orders), nrow(central_orders)),
  mean_sales = c(mean(orders$sales), mean(central_orders$sales)),
  median_sales = c(median(orders$sales), median(central_orders$sales))), digits = 2)
```

scope	n	mean_sales	median_sales
Full frame — primary result	3322	470.80	153.98
Within IQR fences — sensitivity only	3021	257.91	121.44

- The second row changes the scope; it is not a corrected estimate of the original frame mean.
- Large changes show why valid extreme values matter to the business.
- Do not hide exclusions or present the smaller mean as the retailer's full average order value.

7. Independent operations briefing

1. Compare order sales across regions using n, mean, median, SD and IQR.
2. Draw side-by-side boxplots, with the same scale for all regions.

3. Find the region with the largest **total revenue** and the largest **median order**. Are they the same?
4. Identify the top five orders by sales and inspect their product lines.
5. Calculate the share of total revenue contributed by the top 10% of orders ranked by sales. State how you handle ties or rounding of the number of orders.
6. Write a short briefing distinguishing typical size, variability and revenue concentration.

```
# group_by(region) %>% summarise(...) builds the regional table.
# arrange(desc(sales)) ranks order records.
# An explicit rule is to take ceiling(0.10 * nrow(orders)) highest-ranked orders.
```

Your conclusion should include:

- The scope and order-level denominator.
- Two numerical findings with dollar units.
- One useful operational follow-up.
- A limitation: sales values alone do not measure physical workload.

8. Continue

- Practical 3: Discounts and profitability.
- `data/business/README.md`.